DESIGN LAB-1

ME-205

DESIGN AND FABRICATION OF AN UNDERACTUATED TENDON-DRIVEN ROBOTIC GRIPPER

DEPARTMENT OF MECHANICAL ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY ROPAR
RUPNAGAR- 140001, INDIA

THU-GROUP(B)

Sr.no	NAME	ENTRY NUMBER
1	VINIT KUMAR CHOUBEY	2024MEB1400
2	UTKARSH SHUKLA	2024MEB1395
3	RANVEER GOYAL	2024MEB1375
4	SAMEER KUMAR PURBEY	2024MEB1382
5	SHIVAM BADYAL	2024MEB1387

Under the Guidance of: **Dr. Manish Agrawal**

TABLE OF CONTENTS

1. **Project Overview** ....**Page 3**
2. **Working Principle** .... **Page 4**
3. **Mechanical Design and Mechanism** ....**Page 5**
   - 3.1 Underactuated Finger Motion
   - 3.2 Actuation and Transmission (Radial Attachment)
   - 3.3 Passive Return Using Springs
   - 3.4 Speed and Torque Tuning
4. **Under-Actuation and Adaptive Grasping** ....**Page 7**
5. **Electronics and IoT Integration** ....**Page 8**
6. **Comparative Analysis** ....**Page 10**
7. **References and Sources** ....**Page 11**

ABSTRACT

This project is about designing and building the *Tri-Tendon Gripper*, an underactuated, string-driven robotic hand developed at IIT Ropar. Instead of using many motors, we kept the system simple by using just three MG996R servo motor motors to control a total of nine degrees of freedom.

The gripper can automatically adjust its shape while holding objects, so it doesn't need complicated programming for each joint. It uses a combination of active pulling (through tendons) to close the fingers and a passive spring system to open them again, based on Hooke's Law.

We controlled the system using an Arduino Uno R4 WiFi along with the Blynk IoT platform, which allowed us to operate the gripper remotely. Overall, this design is lightweight, energy-efficient, and much simpler compared to traditional robotic hands that rely on complex mechanisms and multiple motors.

1. Project Overview

Our team designed and prototyped a three-finger robotic gripper based on a string-actuated (tendon-driven) mechanism. The idea was to create a lightweight and efficient system that can perform complex finger movements without relying on many motors.

The final design, which we call the *Tri-Tendon Gripper*, uses a 9-degree-of-freedom **(9-DOF)** finger structure but is controlled using only three main tendon lines. Each tendon drives multiple joints in a finger, making the system underactuated.

This approach allows the gripper to automatically adapt to different object shapes while keeping the design simple, compact, and energy-efficient compared to traditional multi-motor systems.

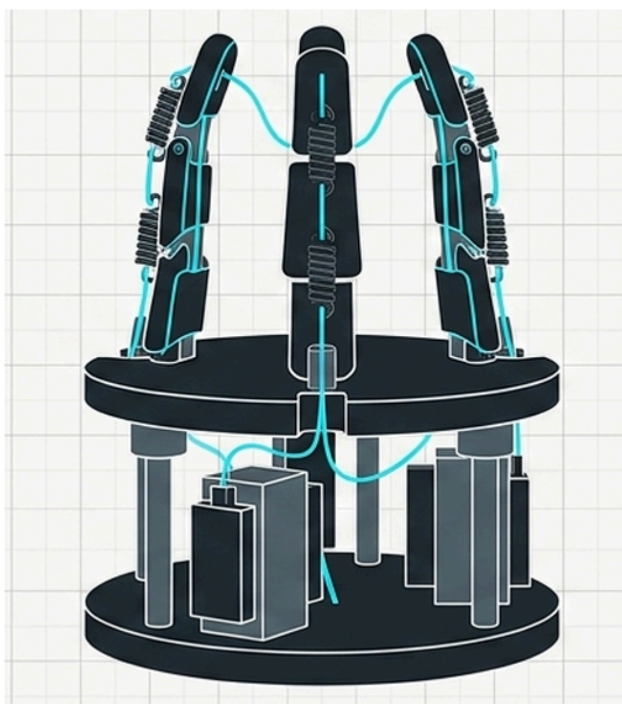

Fig: String actuated gripper model

2. Working Principle

Since our nylon strings can only pull and not push , we had to brainstorm a way to get the fingers to actually open back up once the motor lets go. We went with a "passive" system using springs, which kept the project low-cost and much easier to build.

Instead of using anything complicated, we built normal compression springs directly into the finger joints.

Storing Energy: When our high-torque motors pull the tendons to close the hand, the strings pull against these springs and "squish" them down. This process stores up what is called elastic potential energy inside the spring coils.

The Reset: The moment the motor relaxes the string, all that stored energy is released. The springs push back out to their original shape, which snaps the fingers back into a straight, open position.

The Science: This whole process follows Hooke’s Law. it means that the Force the spring pushes back with is equal to the Stiffness of the spring multiplied by the Distance it was squished. This was a clever trick for our group because it meant we only needed half as many motors, which made our wiring and our Arduino code way easier to manage.

3. Mechanical Design and Mechanism

3.1 Underactuated Finger Motion

- Our gripper uses a 3-finger underactuated design, where each finger has multiple joints but is controlled by a single tendon.
- Instead of fixing exact joint angles, the fingers move depending on how they touch the object.
- This helps the fingers naturally wrap around objects, making the grip more adaptive and reliable.

3.2 Actuation and Transmission (Radial Attachment)

- We used MG996R servo motor motors, with tendons attached to holes on a circular servo horn instead of a spool.
- As the servo rotates, the tendon attachment point moves in a circular path, pulling the tendon.
- The tendon movement depends on the servo angle and radius: $d=r\sin(\theta)$
- This setup gives higher torque at the start of motion, which helps overcome initial resistance in the system.

3.3 Passive Return Using Springs

- Since tendons can only pull, we added compression springs to bring the fingers back to the open position.
- While closing, the motor expands the springs and stores energy in them.

$$E_p=0.5*k(\Delta x)^2$$

- When the motor releases, the springs come back to original position and help reopen the fingers automatically.

3.4 Speed and Torque Tuning

- Because the springs resist motion during closing, we adjusted the servo speeds differently for each action.
- Grip (85): slower movement to provide more torque and avoid damaging the tendon or stalling the motor.
- Retract (105): faster movement to quickly release the tendon while springs assist in opening.
- This tuning made the system smoother, more stable, and easier to control.

For a continuous servo, 90 is the neutral value → the motor stops.

Values **below 90** → rotate in one direction.

Values **above 90** → rotate in the opposite direction.

To optimize performance, our team focused on reducing the weight of the moving parts and improving the structural integrity of the hand.

Design Advantages

- The servos are placed at the base instead of on the fingers, which reduces moving mass and makes the system faster and more efficient.
- Using tendons and springs adds flexibility, so the gripper can handle small impacts without damaging the motor gears.
- This flexible setup acts like a natural shock absorber, making the system safer and more durable.

- It also helps the fingers adjust easily to different object shapes, improving grip without needing complex control

4. Under-Actuation and Adaptive Grasping

The defining feature of this project is **under-actuation**—using only three motors to control nine individual joints.

The Curling Sequence: We achieved a natural "hooking" motion by varying the spring stiffness across the finger. By placing stiffer springs at the base and more flexible springs at the tips, the finger naturally curls from the tip inward, ensuring a more secure wrap around the object.

Adaptive Grasping Capability

- Our tendons are not tied to just one joint, so the fingers can adjust themselves automatically.
- If one part of a finger touches the object early, the tendon keeps pulling the other joints.
- This makes the finger wrap naturally around the object’s shape instead of stopping midway.
- Because of this, the gripper can handle different shapes and sizes easily without extra control.

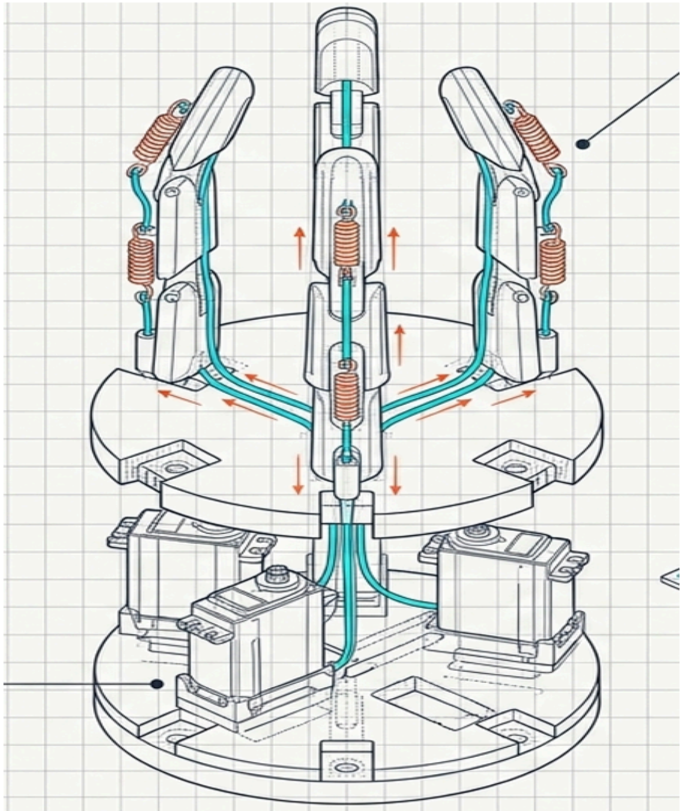

Fig: anatomy of a gripper system

5. Electronics and IoT Integration

The system is powered by the **Arduino Uno R4 WiFi**, which handles the logic for both local and remote control.

- **IoT Interface:** We used the **Blynk IoT platform** for real-time mobile control. We assigned Virtual Pins to represent the finger positions. When a slider is moved on the phone app, the Arduino receives the data and converts it into movement signals for the servos.
- **Power Management:** A major challenge was the "current spikes" from the motors, which can reset the Arduino. We solved this by adding an external

Buck Converter. This device acts as a steady regulator, keeping the motor power at a constant 6.0 Volts while isolating it from the Arduino's sensitive logic circuits.

- **Collision Prevention:** To prevent the fingers from crashing into each other, we wrote a "staggered" control script. Instead of moving all fingers at once, the code introduces a tiny delay between each one: **Start Time of Finger 2 = Start Time of Finger 1 + Time Delay** This also prevents the power system from being overloaded by all three motors starting at the exact same millisecond.

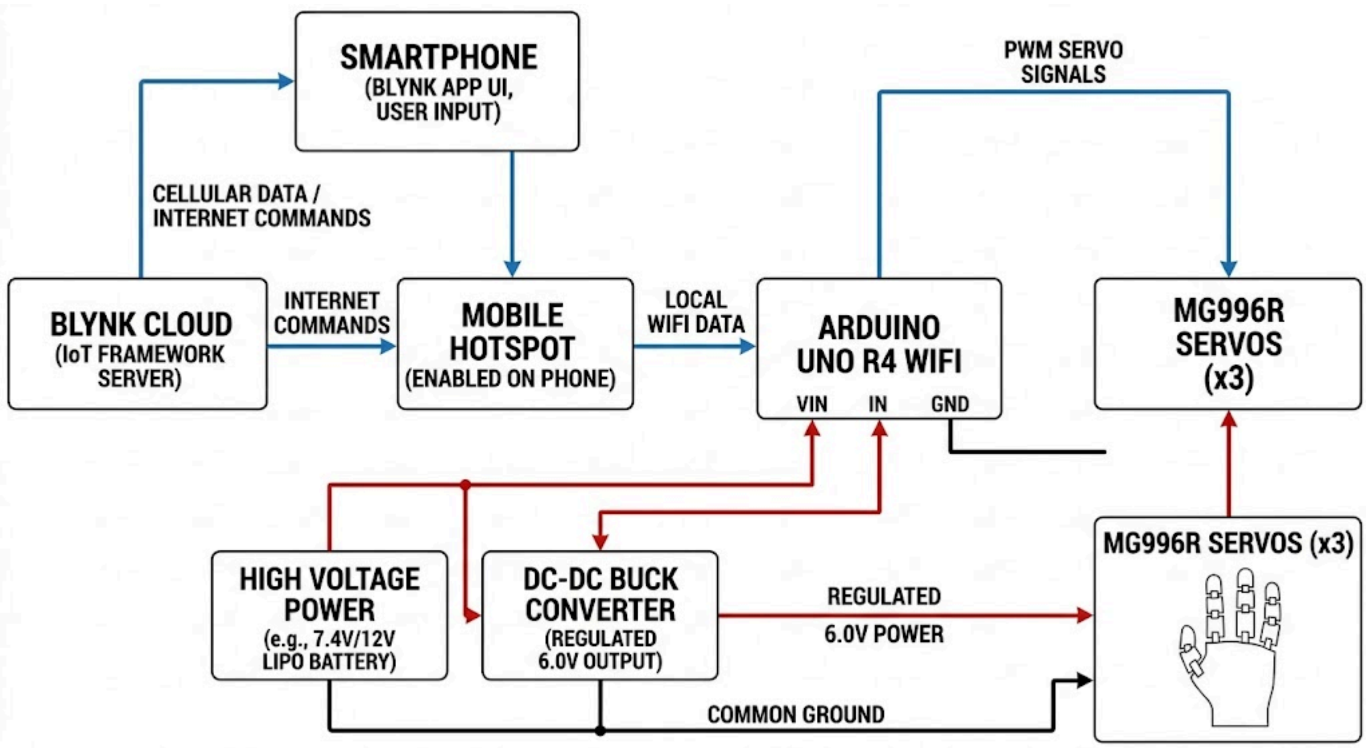

Fig: system architecture

6. Comparative Analysis

The following table summarizes why the Tri-Tendon design was chosen over traditional linkage-based systems.

Feature	Our Tri-Tendon Gripper	Regular Robotic Grippers
Weight	Very light fingers (motors stay in the palm).	Heavy fingers (motors are on the joints).
Shape	Wraps around any shape	Only moves in one fixed way.
Motor Count	Only 3 motors for 9 joints.	Usually needs 9+ motors to do the same thing.

7. References and sources

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